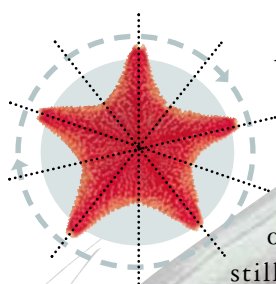




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Shapes with rotational symmetry repeat themselves if you rotate them



Are starfish symmetrical?

A starfish has five axes of lateral symmetry. It also has what mathematicians call *rotational symmetry*, which means you see the same shape repeated if you turn the object round while keeping the centre point still. A parallelogram and the letters N, S, and Z all have rotational symmetry but not lateral symmetry.

Leonardo da Vinci wrote everything backwards in his mirror

Why do mirrors flip the world?

Why does a mirror swap left and right but not top and bottom?

In fact, a mirror doesn't really swap left and right at all.

If you stand in front of a mirror and wave your left hand, the hand that waves back is still on the left.

It isn't a reflection of your right hand, it's your *apparent* left hand opposite your *actual* left hand, just as your apparent head is opposite your actual head. The confusion happens because we imagine ourselves standing behind the mirror. Try to hold a 90° mirror (bottom left) sideways – wow! Now you're upside down!



THINGS TO DO

Mirror writing

Hold some paper on your forehead and write your name on it. Many people write a reflection of their name when they do this, even though mirror writing is normally difficult. The artist Leonardo da Vinci always wrote in mirror writing so that his secret notes were difficult to read.

Make a paper chain



To make a chain of symmetrical figures, fold a long strip of paper in a zigzag and draw half the red man on top, making sure arms and legs extend to the edge. Cut through all the layers and open out the chain.

Palindromes

Sentences that read the same forwards or backwards are called palindromes. "Madam I'm Adam" is a palindrome. Numbers can also be palindromes, and there's a clever way to make them. Take any number with more than one digit, reverse it, then add the two numbers. If you don't get a palindrome first time, repeat the process. Most numbers only take a few steps to make a palindrome, but the numbers 89 and 98 take 24 steps each. Oddly, it's impossible to make a palindrome from 196.

DO THE OBJECTS BELOW HAVE?

